Training Day 13

1. Topics Covered and Brief Explanation

1. Introduction to Classification

Classification is a supervised machine learning technique used to predict categories or classes based on input data. It is used when the output is a predefined category rather than a continuous numerical value.

Examples include spam detection, disease prediction and sentiment analysis.

2. Logistic Regression

Logistic Regression is a supervised learning algorithm used mainly for classification problems. It estimates the probability that an input belongs to a particular category.

Examples:

- Email spam or non-spam classification.
- Predicting whether a customer will purchase a product.
- Identifying positive or negative outcomes.

3. K-Nearest Neighbors (KNN)

KNN is a supervised machine learning algorithm that classifies new data points based on the categories of their nearest neighbouring data points.

It calculates the distance between data points and assigns the class that occurs most frequently among the selected neighbours.

4. Working Principle of KNN

- Select the value of K (number of neighbours).
- Calculate distances between the new data point and existing points.
- Identify the K nearest neighbours.
- Assign the most common class among those neighbours.

5. Applications of Classification Algorithms

- Email spam detection.
- Image classification.
- Customer behaviour prediction.
- Medical diagnosis support.
- Fraud detection.

2. Tools and Technologies Used

Tool	Purpose
Python	Writing classification programs.
Scikit-learn	Implementing Logistic Regression and KNN.
NumPy	Handling numerical data.
Pandas	Organizing datasets.
Google Colab	Executing Python programs.

3. Practical Learning / Code Example

Example: Classification using K-Nearest Neighbors

```
from sklearn.neighbors import KNeighborsClassifier
# Training data
X = [[1, 2], [2, 3], [3, 3] [6, 7], [7, 8], [8, 8]]
y = ["A", "A", "A", "B", "B", "B"]
# Create KNN model
model = KNeighborsClassifier(n_neighbors=3)
# Train the model
model.fit(X, y)
# Predict a new data point
result = model.predict([[2, 2]])
print("Predicted Class:", result[0])
```

Explanation:

- KNeighborsClassifier() creates a KNN classification model.
- n_neighbors=3 selects three nearest neighbours.
- fit() trains the model using the sample dataset.
- predict() determines the category of a new data point.

4. Learning Outcomes

- Understood the basic concept of classification in Machine Learning.
- Learned the working principles of Logistic Regression and KNN.
- Understood how KNN uses neighbouring data points for classification.
- Identified real-world applications of classification algorithms.